

FINAL REPORT

CMSE 344

GROUP NO : 16

PROJECT NAME : Sports Tournament Database Management System

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PROJECT END DATE : May 18rd 2026

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ABSTRACT

This project aims to design a comprehensive Sports Tournament Database Management System to reduce the heavy manual workload and disorganized data often found in managing football events. To achieve this, a strong relational database was developed. The methodology included designing structured tables for teams, players, matches, and venues, alongside implementing a secure, multi-tier user access system (admin, employee, referee, viewer). As a result, the system successfully automates complex tasks, such as tracking live match scores, updating player statistics, and generating real-time tournament standings without data overlap. In conclusion, this project demonstrates that a well-structured digital database significantly reduces administrative errors and provides a reliable, scalable foundation for efficiently managing modern sports tournaments.

Keywords: Database Management, Sports Tournaments, Relational Database, Automated Standings, Role-Based Access.

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1. INTRODUCTION

In this project, we focused on building and designing a strong, modern, and user-friendly website. It provides all the needed information, details, and statistics for football lovers, to stay up-to-date with their favorite team in any tournaments that team participate in. And for the administrators managing the football events. The system comes with extensive venue profiles to show the user all important details about the stadiums and arenas including seating capacities and surface information.

The system will contribute to achieving this purpose by being an innovative user-friendly and well-structured platform that helps employees and match officials through several creative futures that it has.

The project didn't miss being a part of the future by using dynamic real time tracking for detailed player statistics (like goals and assists), automated points tables for tournament standings, and full records for team performances and referee certifications.

2. PROJECT PLANNING AND MANAGEMENT

2.1 Preliminary Project Information

Project Information

Project No	06
Project Name	Sports Tournament Database Management System
Start Date	April 20th 2026
End Date	May 18rd 2026
Time	1 month

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3. DATABASE

3.1 ER Diagram

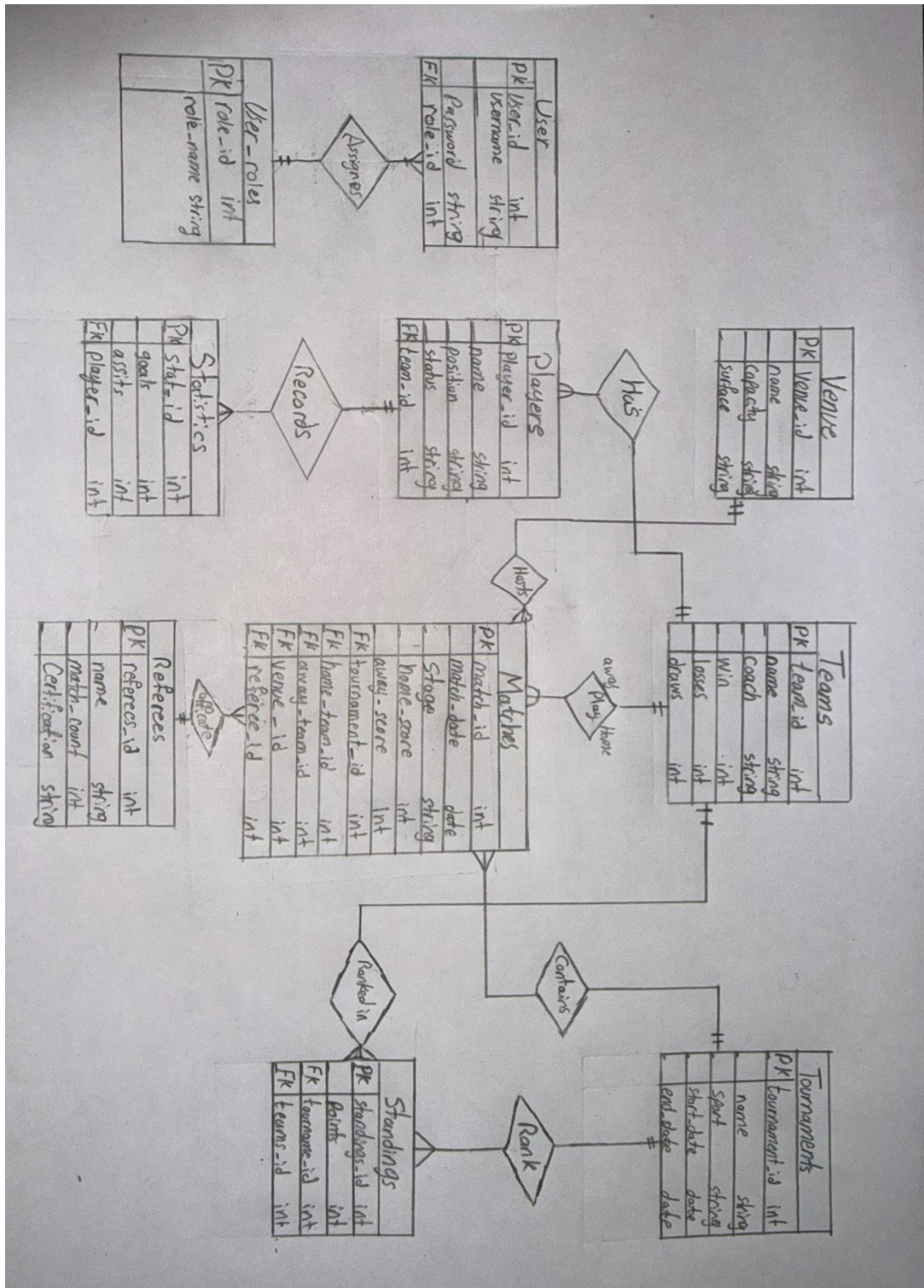


Figure 1 ER Diagram

3.2 Data Definition Language Details (DDL)

3.2.1 User table

```
-- Stores login credentials and role assignment for each system user
CREATE TABLE users (
    user_id      INT AUTO_INCREMENT PRIMARY KEY,
    username     VARCHAR(100) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    email        VARCHAR(150) NOT NULL UNIQUE,
    role_id      INT NOT NULL,
    CONSTRAINT fk_users_role FOREIGN KEY (role_id) REFERENCES user_roles(role_id)
);
```

Figure 2 DDL User Table

3.2.2 User roles table

```
-- Defines the four access levels: admin, employee, referee_user, and viewer
CREATE TABLE user_roles (
    role_id      INT AUTO_INCREMENT PRIMARY KEY,
    role_name     VARCHAR(50) NOT NULL UNIQUE,
    description   VARCHAR(255)
);
```

Figure 3 DDL User Roles Table

3.2.3 Venus table

```
-- Stadiums and arenas where matches are held
CREATE TABLE venues (
    venue_id VARCHAR(10) PRIMARY KEY,
    name      VARCHAR(150) NOT NULL,
    city      VARCHAR(100) NOT NULL,
    country   VARCHAR(100) NOT NULL,
    capacity  INT           NOT NULL CHECK (capacity > 0),
    surface   VARCHAR(50)  NOT NULL
);
```

Figure 4 DDL Venues Table

3.2.4 tournaments table

```
-- Football tournaments with dates, status and prize pool
CREATE TABLE tournaments (
    tournament_id VARCHAR(10) PRIMARY KEY,
    name           VARCHAR(150) NOT NULL,
    start_date     DATE          NOT NULL,
    end_date       DATE          NOT NULL,
    venue_id       VARCHAR(10),
    status         ENUM('Upcoming', 'Ongoing', 'Completed') NOT NULL DEFAULT 'Upcoming',
    prize_pool     BIGINT        NOT NULL DEFAULT 0,
    CONSTRAINT fk_tournament_venue FOREIGN KEY (venue_id) REFERENCES venues(venue_id)
    ON DELETE SET NULL
);
```

Figure 5 DDL Tournaments Table

3.2.5 Teams table

```
-- Participating teams with coach, country and win/loss/draw records
CREATE TABLE teams (
  team_id      VARCHAR(10) PRIMARY KEY,
  name         VARCHAR(100) NOT NULL,
  tournament_id VARCHAR(10) NOT NULL,
  coach        VARCHAR(100),
  country      VARCHAR(100),
  wins         INT NOT NULL DEFAULT 0,
  losses       INT NOT NULL DEFAULT 0,
  draws        INT NOT NULL DEFAULT 0,
  CONSTRAINT fk_team_tournament FOREIGN KEY (tournament_id) REFERENCES tournaments(tournament_id)
  ON DELETE CASCADE
);
```

Figure 6 DDL Teams Table

3.2.6 Referees table

```
-- Match officials with nationality and experience info
CREATE TABLE referees (
  referee_id    VARCHAR(10) PRIMARY KEY,
  name          VARCHAR(100) NOT NULL,
  nationality    VARCHAR(100),
  experience_yrs INT NOT NULL DEFAULT 0 CHECK (experience_yrs >= 0),
  matches_officiated INT NOT NULL DEFAULT 0
);
```

Figure 7 DDL Referees Table

3.2.7 Players table

```
-- Individual athletes with positions, nationality, career goals and assists
• CREATE TABLE players (
    player_id    VARCHAR(10) PRIMARY KEY,
    name         VARCHAR(100) NOT NULL,
    team_id      VARCHAR(10) NOT NULL,
    position     VARCHAR(50),
    nationality   VARCHAR(100),
    age          INT          CHECK (age > 0 AND age < 100),
    jersey       INT          CHECK (jersey >= 0 AND jersey <= 999),
    goals        INT          NOT NULL DEFAULT 0,
    assists      INT          NOT NULL DEFAULT 0,
    status       ENUM('Active','Injured','Suspended','Retired') NOT NULL DEFAULT 'Active',
    CONSTRAINT fk_player_team FOREIGN KEY (team_id) REFERENCES teams(team_id)
        ON DELETE CASCADE
);
```

Figure 8 DDL Players Table

3.2.8 Matches table

```
-- Scheduled and completed game fixtures between two teams
• CREATE TABLE matches (
    match_id     VARCHAR(10) PRIMARY KEY,
    tournament_id VARCHAR(10) NOT NULL,
    home_team_id  VARCHAR(10) NOT NULL,
    away_team_id  VARCHAR(10) NOT NULL,
    venue_id     VARCHAR(10),
    referee_id    VARCHAR(10),
    match_date    DATE        NOT NULL,
    stage         VARCHAR(50),
    home_score    INT          DEFAULT NULL,
    away_score    INT          DEFAULT NULL,
    status        ENUM('Upcoming','Live','Completed','Postponed') NOT NULL DEFAULT 'Upcoming',
    CONSTRAINT fk_match_tournament FOREIGN KEY (tournament_id) REFERENCES tournaments(tournament_id),
    CONSTRAINT fk_match_home_team  FOREIGN KEY (home_team_id) REFERENCES teams(team_id),
    CONSTRAINT fk_match_away_team  FOREIGN KEY (away_team_id) REFERENCES teams(team_id),
    CONSTRAINT fk_match_venue      FOREIGN KEY (venue_id) REFERENCES venues(venue_id) ON DELETE SET NULL,
    CONSTRAINT fk_match_referee    FOREIGN KEY (referee_id) REFERENCES referees(referee_id) ON DELETE SET NULL,
    CONSTRAINT chk_different_teams CHECK (home_team_id <> away_team_id)
);
```

Figure 9 DDL Matches Table

3.2.9 Standings table


```

-- Points table per tournament showing team rankings
CREATE TABLE standings (
    standing_id VARCHAR(10) PRIMARY KEY,
    tournament_id VARCHAR(10) NOT NULL,
    team_id VARCHAR(10) NOT NULL,
    played INT NOT NULL DEFAULT 0,
    won INT NOT NULL DEFAULT 0,
    drawn INT NOT NULL DEFAULT 0,
    lost INT NOT NULL DEFAULT 0,
    goals_for INT NOT NULL DEFAULT 0,
    goals_against INT NOT NULL DEFAULT 0,
    points INT NOT NULL DEFAULT 0,
    position INT NOT NULL DEFAULT 0,
    CONSTRAINT fk_standing_tournament FOREIGN KEY (tournament_id) REFERENCES tournaments(tournament_id) ON DELETE CASCADE,
    CONSTRAINT fk_standing_team FOREIGN KEY (team_id) REFERENCES teams(team_id) ON DELETE CASCADE,
    CONSTRAINT uq_standing UNIQUE (tournament_id, team_id)
);

```

Figure 10 DDL Standings Table

3.2.10 Statistics table

```

-- Player performance stats per individual match (goals, assists, minutes played)
CREATE TABLE statistics (
    stat_id VARCHAR(10) PRIMARY KEY,
    match_id VARCHAR(10) NOT NULL,
    player_id VARCHAR(10) NOT NULL,
    goals INT NOT NULL DEFAULT 0,
    assists INT NOT NULL DEFAULT 0,
    minutes_played INT NOT NULL DEFAULT 0 CHECK (minutes_played >= 0 AND minutes_played <= 120),
    CONSTRAINT fk_stat_match FOREIGN KEY (match_id) REFERENCES matches(match_id) ON DELETE CASCADE,
    CONSTRAINT fk_stat_player FOREIGN KEY (player_id) REFERENCES players(player_id) ON DELETE CASCADE,
    CONSTRAINT uq_stat UNIQUE (match_id, player_id)
);

```

Figure 11 DDL Statistics Table

3.3 Data Manipulation Language (DML)

3.3.1 User table

```

INSERT INTO users (username, password_hash, email, role_id) VALUES
('admin_john', '$2b$10$examplehashADMIN', 'john@tournamentdb.com', 1),
('emp_sara', '$2b$10$examplehashEMP1', 'sara@tournamentdb.com', 2),
('referee_marc', '$2b$10$examplehashREF1', 'marc@tournamentdb.com', 3),
('viewer_ali', '$2b$10$examplehashVIEW1', 'ali@tournamentdb.com', 4);

```

Figure 12 DML User Table

3.3.2 User roles table

```
INSERT INTO user_roles (role_name, description) VALUES
('admin', 'Full access to all tables and system settings'),
('employee', 'Can insert and update match results and standings'),
('referee_user', 'Can view assigned matches and submit reports'),
('viewer', 'Read-only access to public tournament data');
```

Figure 13 DML User Roles Table

3.3.3 Venus table

```
INSERT INTO venues VALUES
('v1', 'MetLife Stadium', 'New York', 'USA', 82500, 'Grass'),
('v2', 'Wembley Stadium', 'London', 'UK', 90000, 'Grass'),
('v3', 'Estadio Azteca', 'Mexico City', 'Mexico', 87523, 'Grass'),
('v4', 'Camp Nou', 'Barcelona', 'Spain', 99354, 'Grass'),
('v5', 'Allianz Arena', 'Munich', 'Germany', 75000, 'Grass'),
('v6', 'San Siro', 'Milan', 'Italy', 80018, 'Grass');
```

Figure 14 DML Venus Table

3.3.4 tournaments table

```
INSERT INTO tournaments VALUES
('t1', 'World Cup 2026', '2026-06-10', '2026-07-11', 'v1', 'Upcoming', 500000000),
('t2', 'Champions League 25/26', '2025-09-16', '2026-05-30', 'v4', 'Ongoing', 2100000000),
('t3', 'Europa League 25/26', '2025-09-18', '2026-05-20', 'v6', 'Ongoing', 600000000);
```

Figure 15 DML tournaments Table

3.3.5 Teams table

```
INSERT INTO teams VALUES
('tm1', 'Brazil',      't1', 'Ancelotti', 'Brazil', 0, 0, 0),
('tm2', 'Germany',    't1', 'Nagelsmann', 'Germany', 0, 0, 0),
('tm3', 'France',     't1', 'Deschamps', 'France', 0, 0, 0),
('tm4', 'Real Madrid', 't2', 'Ancelotti', 'Spain', 4, 1, 1),
('tm5', 'Man City',    't2', 'Guardiola', 'England', 3, 1, 2),
('tm6', 'Ajax',       't3', 'Farioli', 'Netherlands', 3, 1, 1),
('tm7', 'Roma',       't3', 'Ranieri', 'Italy', 2, 2, 1);
```

Figure 16 DML Teams Table

3.3.6 Referees table

```
INSERT INTO referees VALUES
('r1', 'Szymon Marciniak', 'Polish', 15, 210),
('r2', 'Slavko Vincic', 'Slovenian', 12, 165),
('r3', 'Anthony Taylor', 'English', 14, 198),
('r4', 'Felix Brych', 'German', 18, 280),
('r5', 'Clement Turpin', 'French', 13, 175);
```

Figure 17 DML Referees Table

3.3.7 Players table

```
INSERT INTO players VALUES
('p1', 'Vinicius Junior', 'tm1', 'Forward', 'Brazilian', 24, 7, 18, 12, 'Active'),
('p2', 'Rodrygo', 'tm1', 'Forward', 'Brazilian', 23, 11, 9, 8, 'Active'),
('p3', 'Florian Wirtz', 'tm2', 'Midfielder', 'German', 21, 10, 15, 18, 'Active'),
('p4', 'Kyllian Mbappe', 'tm3', 'Forward', 'French', 26, 10, 31, 11, 'Active'),
('p5', 'Jude Bellingham', 'tm4', 'Midfielder', 'English', 21, 5, 19, 14, 'Active'),
('p6', 'Erling Haaland', 'tm5', 'Forward', 'Norwegian', 24, 9, 34, 7, 'Active'),
('p7', 'Phil Foden', 'tm5', 'Midfielder', 'English', 25, 47, 16, 22, 'Active'),
('p8', 'Dusan Tadic', 'tm6', 'Midfielder', 'Serbian', 35, 10, 8, 12, 'Active'),
('p9', 'Stephan El Shaarawy', 'tm7', 'Forward', 'Italian', 31, 92, 7, 5, 'Active');
```

Figure 18 DML Players Table

3.3.8 Matches table

```
INSERT INTO matches VALUES
('m1', 't2', 'tm4', 'tm5', 'v4', 'r1', '2025-11-05', 'Group Stage', 3, 2, 'Completed'),
('m2', 't2', 'tm5', 'tm4', 'v2', 'r3', '2025-12-10', 'Group Stage', 1, 1, 'Completed'),
('m3', 't3', 'tm6', 'tm7', 'v6', 'r2', '2025-11-07', 'Group Stage', 2, 0, 'Completed'),
('m4', 't2', 'tm4', 'tm5', 'v4', 'r4', '2026-04-30', 'Semi-Final', NULL, NULL, 'Upcoming'),
('m5', 't3', 'tm7', 'tm6', 'v5', 'r5', '2026-03-12', 'Round of 16', NULL, NULL, 'Upcoming');
```

Figure 19 DML Matches Table

3.3.9 Standings table

```
INSERT INTO standings VALUES
('s1', 't2', 'tm4', 6, 4, 1, 1, 18, 9, 13, 1),
('s2', 't2', 'tm5', 6, 3, 2, 1, 15, 10, 11, 2),
('s3', 't3', 'tm6', 5, 3, 1, 1, 10, 4, 10, 1),
('s4', 't3', 'tm7', 5, 2, 2, 1, 8, 7, 8, 2);
```

Figure 20 DML Standings Table

3.3.10 Statistics table

```
INSERT INTO statistics VALUES
('st1', 'm1', 'p5', 2, 1, 90),
('st2', 'm1', 'p6', 2, 0, 88),
('st3', 'm1', 'p7', 0, 2, 90),
('st4', 'm2', 'p5', 1, 0, 90),
('st5', 'm2', 'p6', 1, 0, 90),
('st6', 'm2', 'p7', 0, 1, 85),
('st7', 'm3', 'p8', 1, 1, 90),
('st8', 'm3', 'p9', 0, 0, 75);
```

Figure 21 DML Statistics Table

5. QUERIES & SQL BLOCKS

5.1 Queries

5.1.1 Query 1 Tournament overview with venue details

```
SELECT

    t.tournament_id,

    t.name      AS tournament_name,

    t.status,

    t.start_date,

    t.end_date,

    v.name      AS venue_name,

    v.city,

    v.country,

    v.capacity,

    FORMAT(t.prize_pool, 0) AS prize_pool

FROM tournaments t

LEFT JOIN venues v ON t.venue_id = v.venue_id

ORDER BY t.start_date;
```

	tournament_id	tournament_name	status	start_date	end_date	venue_name	city	country	capacity	
▶	t3	NBA Finals 2025	Completed	2025-06-05	2025-06-22	Madison Square Garden	New York	USA	20789	2
	t4	Wimbledon 2025	Completed	2025-06-30	2025-07-13	Centre Court Wimbledon	London	UK	14979	5
	t2	Champions League 25/26	Ongoing	2025-09-16	2026-05-30	Camp Nou	Barcelona	Spain	99354	2
	t1	World Cup 2026	Upcoming	2026-06-10	2026-07-11	MetLife Stadium	New York	USA	82500	5

5.1.2 Query 2 Top Scores (with JOIN, ORDER BY, LIMIT)

```
SELECT
```

```

    p.name AS player_name,

    p.position,

    p.nationality,

    p.goals,

    p.assists,

    (p.goals + p.assists) AS goal_contributions,

    tm.name AS team_name

FROM players p

JOIN teams tm ON p.team_id = tm.team_id

WHERE p.status = 'Active'

ORDER BY p.goals DESC

LIMIT 5;

```

Result Grid							
Filter Rows:				Export:		Wrap Cell Content:	
	player_name	position	nationality	goals	assists	goal_contributions	team_name
▶	Erling Haaland	Forward	Norwegian	34	7	41	Man City
	Kylian Mbappe	Forward	French	31	11	42	France
	Jude Bellingham	Midfielder	English	19	14	33	Real Madrid
	Vinicius Junior	Forward	Brazilian	18	12	30	Brazil
	Phil Foden	Midfielder	English	16	22	38	Man City

5.1.3 Query 3 Match Results with CASE Statement

```

SELECT

    m.match_id,

    ht.name AS home_team,

    at.name AS away_team,

```

```

m.home_score,

m.away_score,

CASE

    WHEN m.home_score > m.away_score THEN ht.name

    WHEN m.away_score > m.home_score THEN at.name

    ELSE 'Draw'

END AS winner,

m.match_date,

m.status

FROM matches m

JOIN teams ht ON m.home_team_id = ht.team_id

JOIN teams at ON m.away_team_id = at.team_id

WHERE m.home_score IS NOT NULL

ORDER BY m.match_date DESC;

```

Result Grid								
			Filter Rows:		Export:		Wrap Cell Content:	
	match_id	home_team	away_team	home_score	away_score	winner	match_date	status
▶	m2	Man City	Real Madrid	1	1	Draw	2025-12-10	Completed
	m3	Ajax	Roma	2	0	Ajax	2025-11-07	Completed
	m1	Real Madrid	Man City	3	2	Real Madrid	2025-11-05	Completed

5.1.4 Query 4 Tournament Statistics (with GROUP BY, COUNT, SUM, AVG)

```

SELECT

    t.name AS tournament_name,

    COUNT(DISTINCT tm.team_id) AS total_teams,

```

```

COUNT(DISTINCT p.player_id) AS total_players,

SUM(p.goals) AS total_goals,

AVG(p.goals) AS avg_goals_per_player,

FORMAT(t.prize_pool, 0) AS prize_pool

FROM tournaments t

LEFT JOIN teams tm ON tm.tournament_id = t.tournament_id

LEFT JOIN players p ON p.team_id = tm.team_id

GROUP BY t.tournament_id, t.name, t.prize_pool;

```

	tournament_name	total_teams	total_players	total_goals	avg_goals_per_player	prize_pool
▶	World Cup 2026	3	4	73	18.2500	500,000,000
	Champions League 25/26	2	3	69	23.0000	2,100,000,000
	NBA Finals 2025	2	2	15	7.5000	22,000,000
	Wimbledon 2025	0	0	NULL	NULL	50,000,000

5.1.5 Query 5 Referees Ranked by Experience (with LEFT JOIN, GROUP BY)

```

SELECT

    r.name,

    r.nationality,

    r.experience_yrs,

    r.matches_officiated,

    COUNT(m.match_id) AS matches_in_system,

    ROUND(r.matches_officiated / r.experience_yrs, 1) AS avg_matches_per_year

FROM referees r

```


LEFT JOIN matches m ON m.referee_id = r.referee_id

GROUP BY r.referee_id, r.name, r.nationality, r.experience_yrs, r.matches_officiated

ORDER BY r.experience_yrs DESC;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	name	nationality	experience_yrs	matches_officiated	matches_in_system	avg_matches_per_year
▶	Felix Brych	German	18	280	0	15.6
	Szymon Marciniak	Polish	15	210	1	14.0
	Anthony Taylor	English	14	198	1	14.1
	Clement Turpin	French	13	175	0	13.5
	Slavko Vincic	Slovenian	12	165	1	13.8

5.2 SQL blocks

5.2.1 Procedure

5.2.1.1 Procedure 1 Record match result and auto-update standings

DELIMITER \$\$

CREATE PROCEDURE sp_record_match_result(

IN p_match_id VARCHAR(10),

IN p_home_score INT,

IN p_away_score INT

)

BEGIN

DECLARE v_home_team VARCHAR(10);

DECLARE v_away_team VARCHAR(10);

DECLARE v_tourn VARCHAR(10);

SELECT home_team_id, away_team_id, tournament_id

INTO v_home_team, v_away_team, v_tourn

FROM matches WHERE match_id = p_match_id;

UPDATE matches

SET home_score = p_home_score,

away_score = p_away_score,

status = 'Completed'

WHERE match_id = p_match_id;

IF p_home_score > p_away_score THEN

UPDATE standings SET played = played+1, won = won+1, points = points+3,

goals_for = goals_for+p_home_score, goals_against =
goals_against+p_away_score

WHERE team_id = v_home_team AND tournament_id = v_tourn;

UPDATE standings SET played = played+1, lost = lost+1,

goals_for = goals_for+p_away_score, goals_against =
goals_against+p_home_score

WHERE team_id = v_away_team AND tournament_id = v_tourn;

ELSEIF p_away_score > p_home_score THEN

UPDATE standings SET played = played+1, lost = lost+1,

goals_for = goals_for+p_home_score, goals_against =
goals_against+p_away_score

```

WHERE team_id = v_home_team AND tournament_id = v_tourn;

UPDATE standings SET played = played+1, won = won+1, points = points+3,
    goals_for      =      goals_for+p_away_score,      goals_against      =
goals_against+p_home_score

WHERE team_id = v_away_team AND tournament_id = v_tourn;

ELSE

UPDATE standings SET played = played+1, drawn = drawn+1, points = points+1,
    goals_for      =      goals_for+p_home_score,      goals_against      =
goals_against+p_away_score

WHERE team_id = v_home_team AND tournament_id = v_tourn;

UPDATE standings SET played = played+1, drawn = drawn+1, points = points+1,
    goals_for      =      goals_for+p_away_score,      goals_against      =
goals_against+p_home_score

WHERE team_id = v_away_team AND tournament_id = v_tourn;

END IF;

SELECT 'Match result recorded and standings updated.' AS result;

END$$

```

5.2.1.2 Procedure 2 Get full squad for a team

```

CREATE PROCEDURE sp_get_team_squad(

    IN p_team_id VARCHAR(10)

)

```

```

BEGIN

SELECT

    p.player_id, p.name, p.position, p.nationality,

    p.age, p.jersey, p.goals, p.assists, p.status,

    t.name AS team_name,

    tr.name AS tournament_name

FROM players p

JOIN teams    t ON p.team_id    = t.team_id

JOIN tournaments tr ON t.tournament_id = tr.tournament_id

WHERE p.team_id = p_team_id

ORDER BY p.position, p.jersey;

END$$

```

5.2.2 Function

5.2.2.1 Function 1 Calculate goal difference for a team

```

CREATE FUNCTION fn_goal_difference(

    p_team_id    VARCHAR(10),

    p_tournament_id VARCHAR(10)

)

RETURNS INT

DETERMINISTIC

BEGIN

```

```

DECLARE v_gf INT DEFAULT 0;

DECLARE v_ga INT DEFAULT 0;

SELECT goals_for, goals_against INTO v_gf, v_ga

FROM standings

WHERE team_id = p_team_id AND tournament_id = p_tournament_id;

RETURN (v_gf - v_ga);

END$$

```

5.2.2.2 Function 2 Get total goals + assists for a player in a match

```

CREATE FUNCTION fn_player_match_contribution(

    p_player_id VARCHAR(10),

    p_match_id VARCHAR(10)

)

RETURNS INT

DETERMINISTIC

BEGIN

    DECLARE v_total INT DEFAULT 0;

    SELECT COALESCE(goals + assists, 0)

    INTO v_total

    FROM statistics

    WHERE player_id = p_player_id AND match_id = p_match_id;

    RETURN v_total;

END$$

```

5.2.3 Trigger

5.2.3.1 Trigger 1 Auto-sync teams table when standings are updated

```
CREATE TRIGGER trg_sync_team_record
```

```
AFTER UPDATE ON standings
```

```
FOR EACH ROW
```

```
BEGIN
```

```
    UPDATE teams
```

```
    SET wins  = NEW.won,
```

```
        losses = NEW.lost,
```

```
        draws = NEW.drawn
```

```
    WHERE team_id = NEW.team_id;
```

```
END$$
```

5.2.4 Sample Procedure & Function Calls

- Call Procedure 1: Record match result

```
CALL sp_record_match_result('m3', 2, 0);
```



The screenshot shows a window titled 'Result Grid' with a 'Filter Rows:' input field. Below the header, there is a table with one row containing the text 'Match result recorded and standings updated.'.

result
Match result recorded and standings updated.

Figure 22 Record match result

- Call Procedure 2: Get team squad

```
CALL sp_get_team_squad('tm4');
```

Result Grid											
Filter Rows:											
	player_id	name	position	nationality	age	jersey	goals	assists	status	team_name	tournament_name
▶	p5	Jude Bellingham	Midfielder	English	21	5	19	14	Active	Real Madrid	Champions League

Figure 23 Get team squad

- Call Function 1: Goal difference for Real Madrid in Champions League

```
SELECT fn_goal_difference('tm4', 't2') AS real_madrid_goal_diff;
```

Result Grid		Filter Rows:	
	real_madrid_goal_diff		
▶	9		

Figure 24 Goal difference for Real Madrid in Champions League

- Call Function 2: Bellingham contribution in match m1

```
SELECT fn_player_match_contribution('p5', 'm1') AS
```

```
bellingham_match_contribution;
```

Result Grid		Filter Rows:	
	bellingham_match_contribution		
▶	3		

Figure 25 Bellingham Contribution in Match m1

5.2.5 Trigger Test

- Check BEFORE trigger

```
SELECT team_id, wins, losses, draws FROM teams WHERE team_id = 'tm4';
```

Result Grid				
	team_id	wins	losses	draws
▶	tm4	5	3	1
✱	NULL	NULL	NULL	NULL

Figure 26 Trigger BEFORE

- Update standings (activates trigger)

UPDATE standings SET won = 6, lost = 2, drawn = 1 WHERE team_id = 'tm4';



Figure 27 Trigger UPDATE

- Check AFTER trigger (wins/losses/draws should auto-update)

SELECT team_id, wins, losses, draws FROM teams WHERE team_id = 'tm4';

Output				
#	Time	Action	Message	
1	17:06:15	USE tournamentdb	0 row(s) aff	
2	17:06:16	SELECT team_id, wins, losses, draws FROM teams WHERE team_id = 'tm4' LIMIT 0, 1000	1 row(s) ret	

Figure 28 Trigger AFTER

- Reset to original values

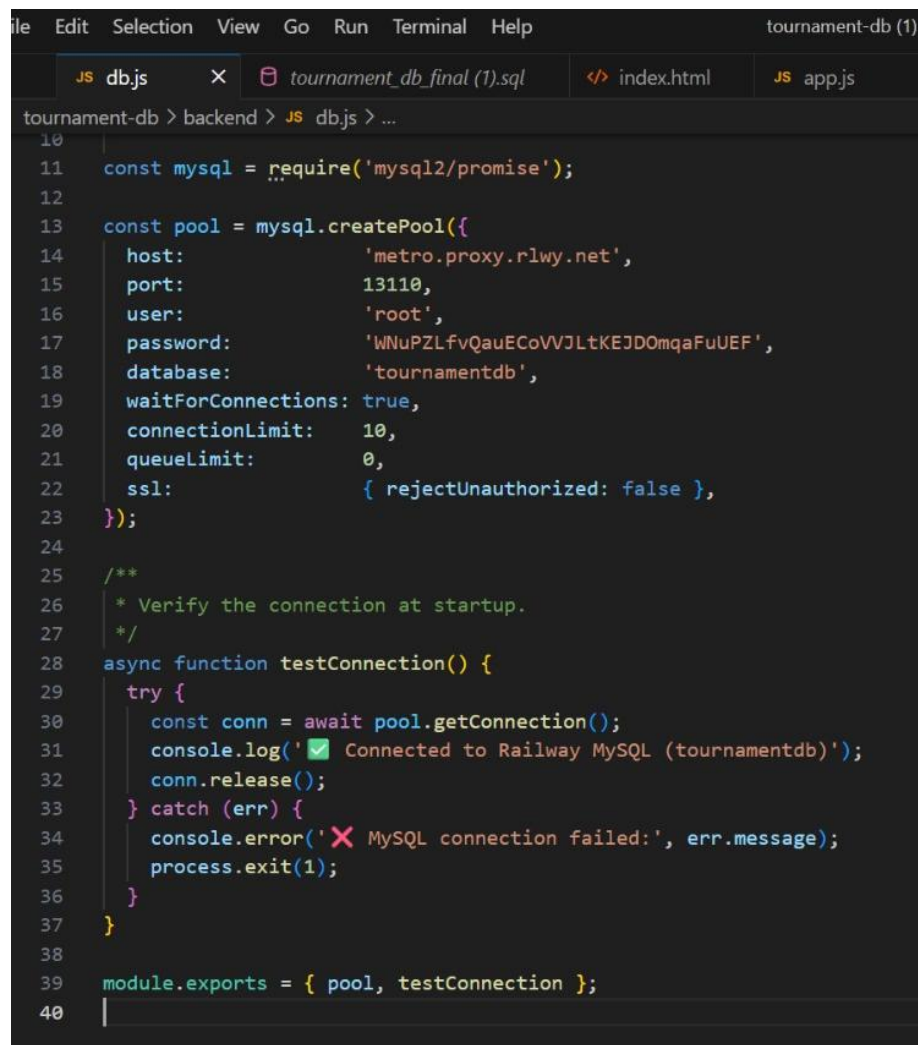

```
UPDATE standings SET won = 4, lost = 1, drawn = 1 WHERE team_id =  
'tm4';
```

6. SOFTWARE CONNECTION & GUI

6.1 Used tools

The website was implemented using Python with FastAPI for the backend, and HTML, CSS, JS for the frontend, and for database we used MySQL.

6.2 Database connection



```
10
11 const mysql = require('mysql2/promise');
12
13 const pool = mysql.createPool({
14   host:      'metro.proxy.rlwy.net',
15   port:      13110,
16   user:      'root',
17   password:  'WNuPZLfVQauECovVJLtKEJD0mqaFuUEF',
18   database:  'tournamentdb',
19   waitForConnections: true,
20   connectionLimit: 10,
21   queueLimit: 0,
22   ssl:      { rejectUnauthorized: false },
23 });
24
25 /**
26  * Verify the connection at startup.
27  */
28 async function testConnection() {
29   try {
30     const conn = await pool.getConnection();
31     console.log('✅ Connected to Railway MySQL (tournamentdb)');
32     conn.release();
33   } catch (err) {
34     console.error('❌ MySQL connection failed:', err.message);
35     process.exit(1);
36   }
37 }
38
39 module.exports = { pool, testConnection };
40
```

Figure 29 Database Linking Connection

6.3 Graphical user interface

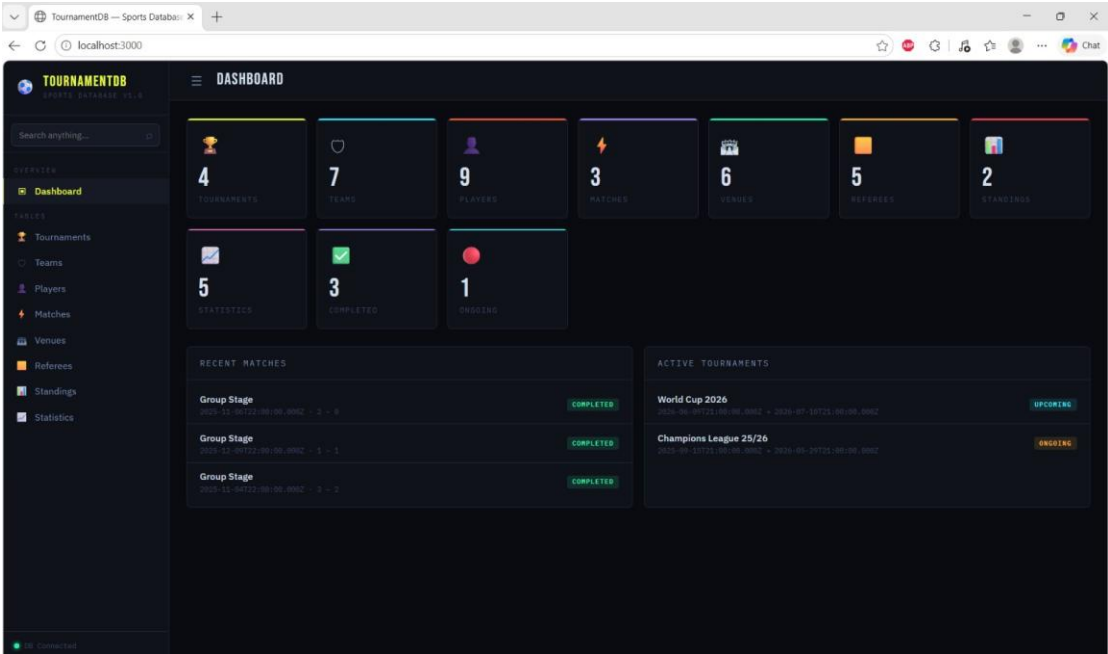


Figure 30 Dashboard Page

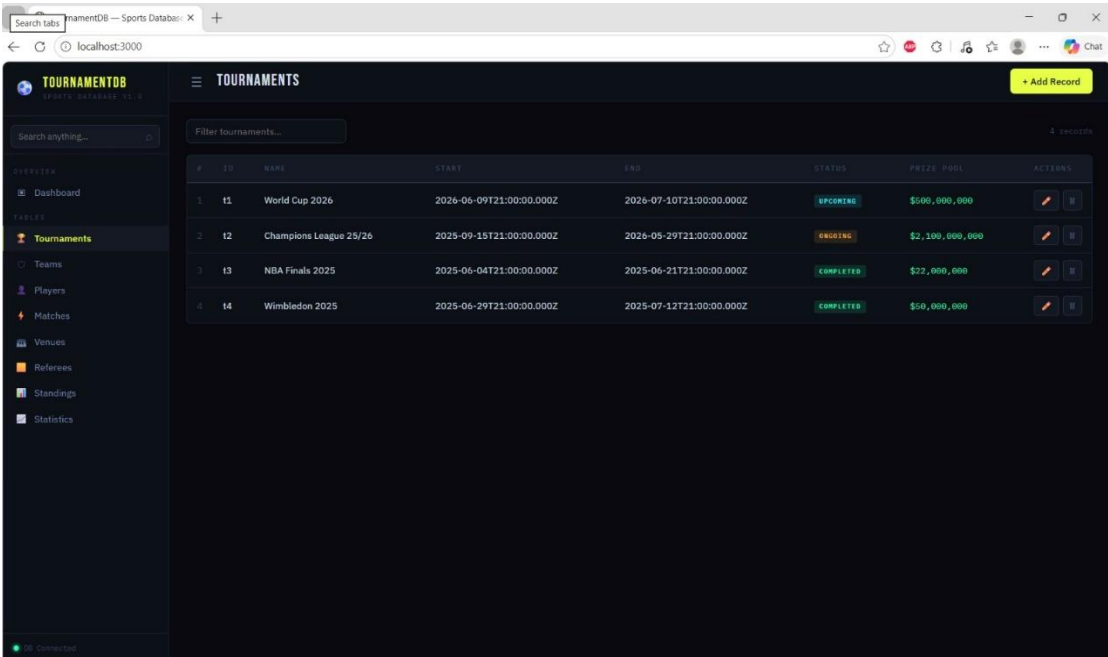


Figure 31 Tournaments Page

TOURNAMENTDB
SPORTS DATABASE V1.0

PLAYERS

+ Add Record

Search anything...

Filter players...

9 records

#	ID	NAME	POSITION	NATIONALITY	AGE	#	GOALS	ASSISTS	STATUS	ACTIONS
1	p1	Vinicius Junior	Forward	Brazilian	24	7	18	12	ACTIVE	
2	p2	Rodrygo	Forward	Brazilian	23	11	9	8	ACTIVE	
3	p3	Florian Wirtz	Midfielder	German	21	10	15	18	ACTIVE	
4	p4	Kylian Mbappe	Forward	French	26	10	31	11	ACTIVE	
5	p5	Jude Bellingham	Midfielder	English	21	5	19	14	ACTIVE	
6	p6	Erling Haaland	Forward	Norwegian	24	9	34	7	ACTIVE	
7	p7	Phil Foden	Midfielder	English	25	47	16	22	ACTIVE	
8	p8	Dusan Tadic	Midfielder	Serbian	35	10	8	12	ACTIVE	
9	p9	Stephan El Shearawy	Forward	Italian	31	92	7	5	ACTIVE	

DB Connected

Figure 32 Players Page

TOURNAMENTDB
SPORTS DATABASE V1.0

MATCHES

+ Add Record

Search anything...

Filter matches...

3 records

#	ID	DATE	STAGE	HOME	AWAY	H	A	STATUS	ACTIONS
1	m1	2025-11-04T22:00:00.000Z	Group Stage	tm4	tm5	3	2	COMPLETED	
2	m2	2025-12-09T22:00:00.000Z	Group Stage	tm5	tm4	1	1	COMPLETED	
3	m3	2025-11-06T22:00:00.000Z	Group Stage	tm6	tm7	2	0	COMPLETED	

DB Connected

Figure 33 Matches Page

7. CONCLUSION

In conclusion, this project has successfully delivered a strong and highly efficient Sports Tournaments Database Management Systems was carefully designing the database and handling everything from complex match fixtures and referee assignments to venue capacities, we created a centralized platform that brings order to managing football tournaments.

The system we made achieves its main goal by significantly reducing the manual workload for administrators, while providing accurate, accessible information for viewers.

This project proved its readiness for the future by providing a scalable and secure technical foundation. As we hope to see in the future tournaments grow and more teams, players, or matches added, the database is fully equipped to remain a reliable, and user-friendly solution that elevates the standard of sports management.